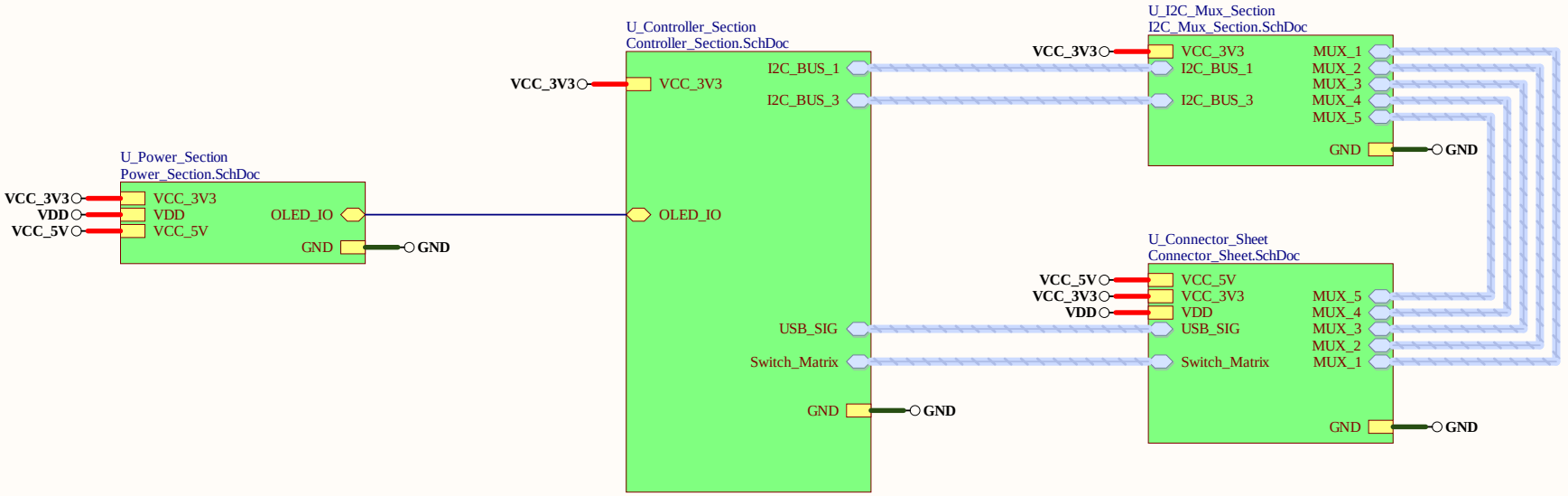
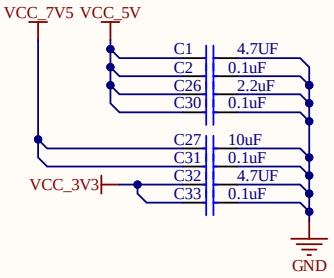
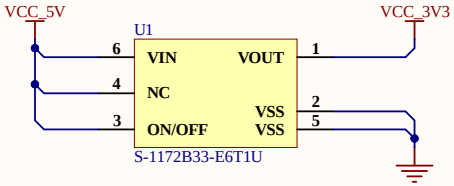

U_Power_Section
Power_Section.SchDoc
VCC_3V3
VDD
VCC_5V
VCC_3V3
VDD
OLED_IO
GND
GND
U_Controller_Section
Controller_Section.SchDoc
VCC_3V3
VCC_3V3
I2C_BUS_1
I2C_BUS_3
OLED_IO
USB_SIG
Switch_Matrix
GND
GND
U_I2C_Mux_Section
I2C_Mux_Section.SchDoc
VCC_3V3
VCC_3V3
I2C_BUS_1
I2C_BUS_3
MUX_1
MUX_2
MUX_3
MUX_4
MUX_5
GND
GND
U_Connector_Sheet
Connector_Sheet.SchDoc
VCC_5V
VCC_3V3
VDD
VCC_5V
VCC_3V3
VDD
USB_SIG
Switch_Matrix
MUX_5
MUX_4
MUX_3
MUX_2
MUX_1
GND
GND

OLED_IO OLED_IO

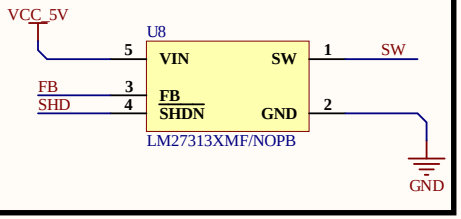
DECOUPLING CAPACITORS



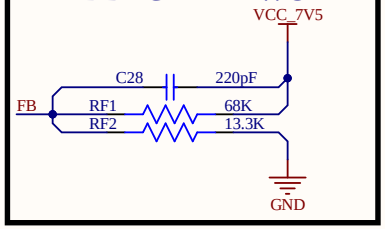
3.3V FIXED REGULATOR



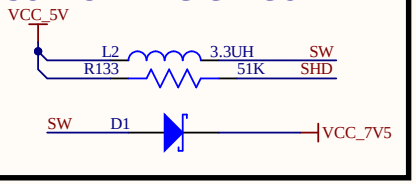
BOOST REGULATOR



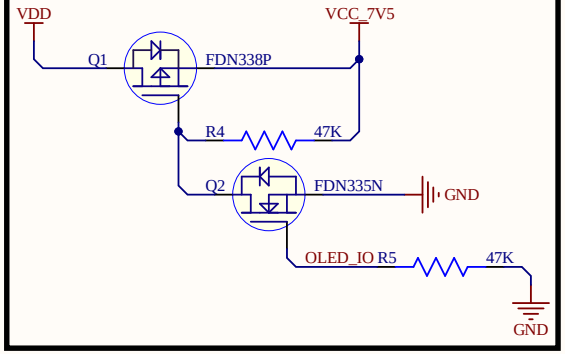
FEEDBACK NETWORK

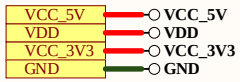


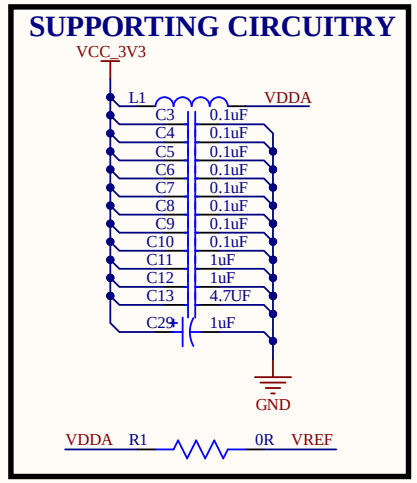
SUPPORTING CIRCUIT

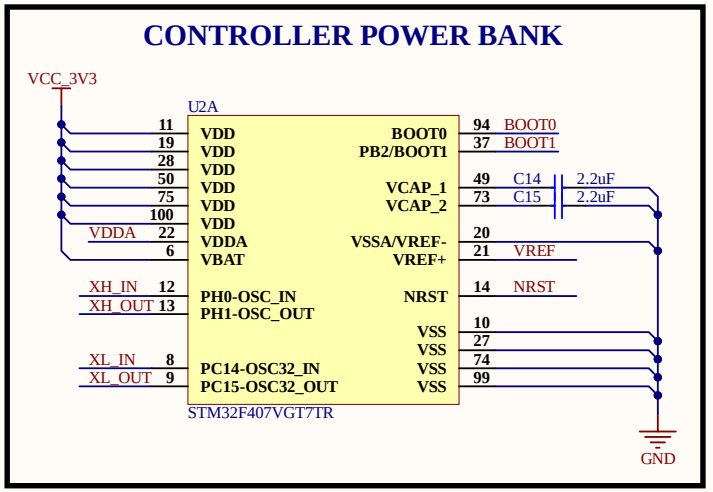


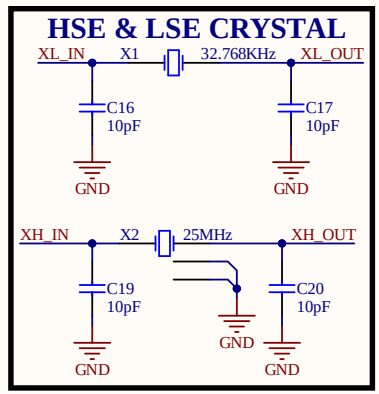
MOSFET CONTROL CIRCUIT

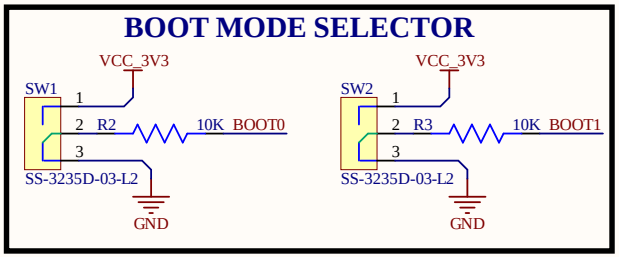


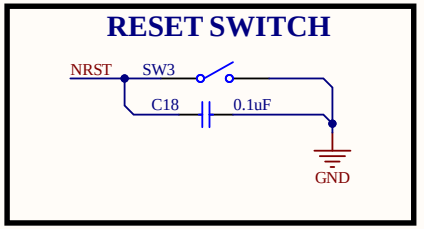



SUPPORTING CIRCUITRY
VCC_3V3
L1
VDDA
C3 0.1uF
C4 0.1uF
C5 0.1uF
C6 0.1uF
C7 0.1uF
C8 0.1uF
C9 0.1uF
C10 0.1uF
C11 1uF
C12 1uF
C13 4.7UF
C29 1uF
GND
VDDA R1 0R VREF

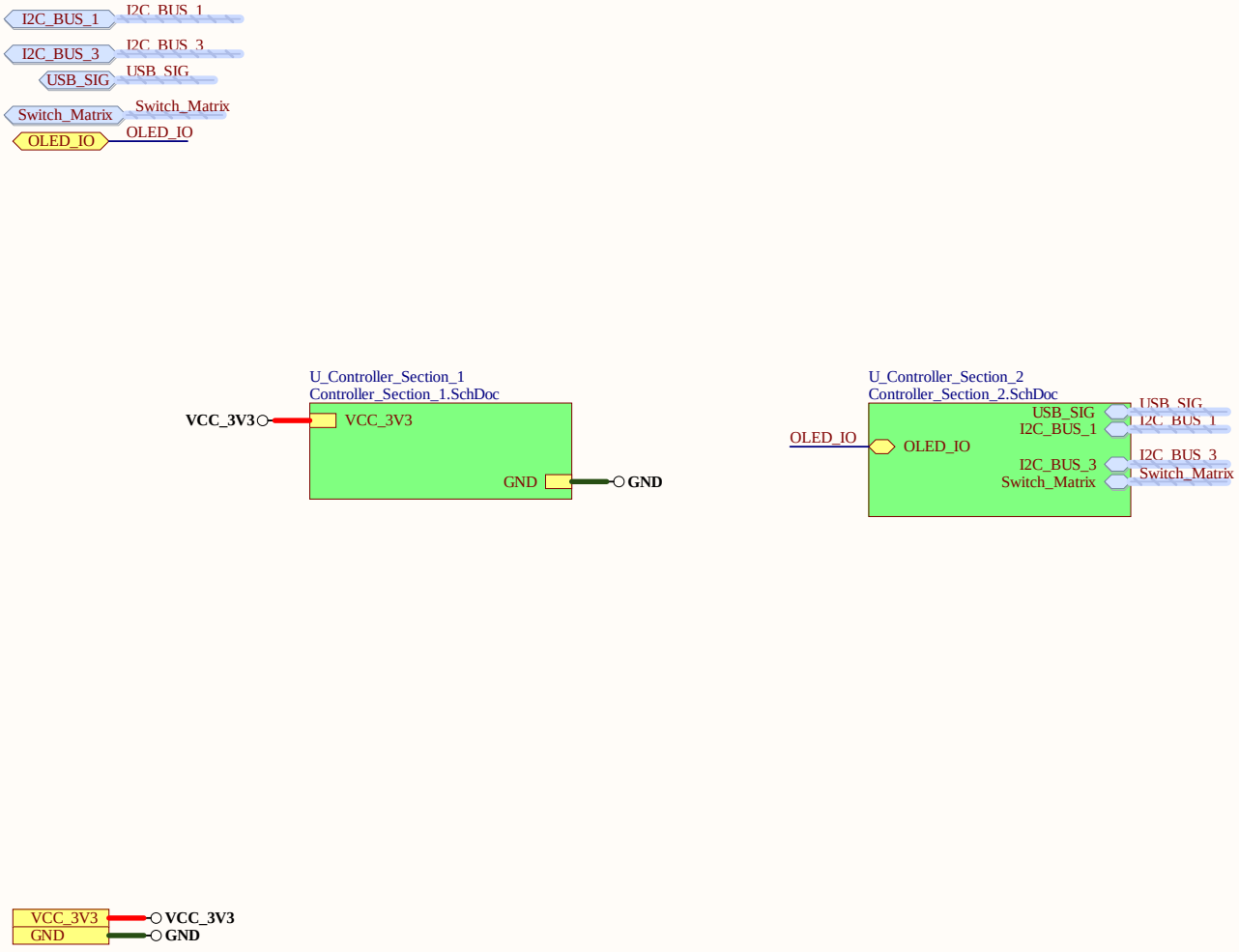

CONTROLLER POWER BANK
VCC_3V3
U2A
11 VDD
19 VDD
28 VDD
50 VDD
75 VDD
100 VDD
VDDA 22 VDDA
6 VBAT
XH_IN 12 PH0-OSC_IN
XH_OUT 13 PH1-OSC_OUT
XL_IN 8 PC14-OSC32_IN
XL_OUT 9 PC15-OSC32_OUT
BOOT0 94 BOOT0
PB2/BOOT1 37 BOOT1
VCAP_1 49 C14 2.2uF
VCAP_2 73 C15 2.2uF
VSSA/VREF- 20
VREF+ 21 VREF
NRST 14 NRST
VSS 10
VSS 27
VSS 74
VSS 99
STM32F407VGT7TR
GND

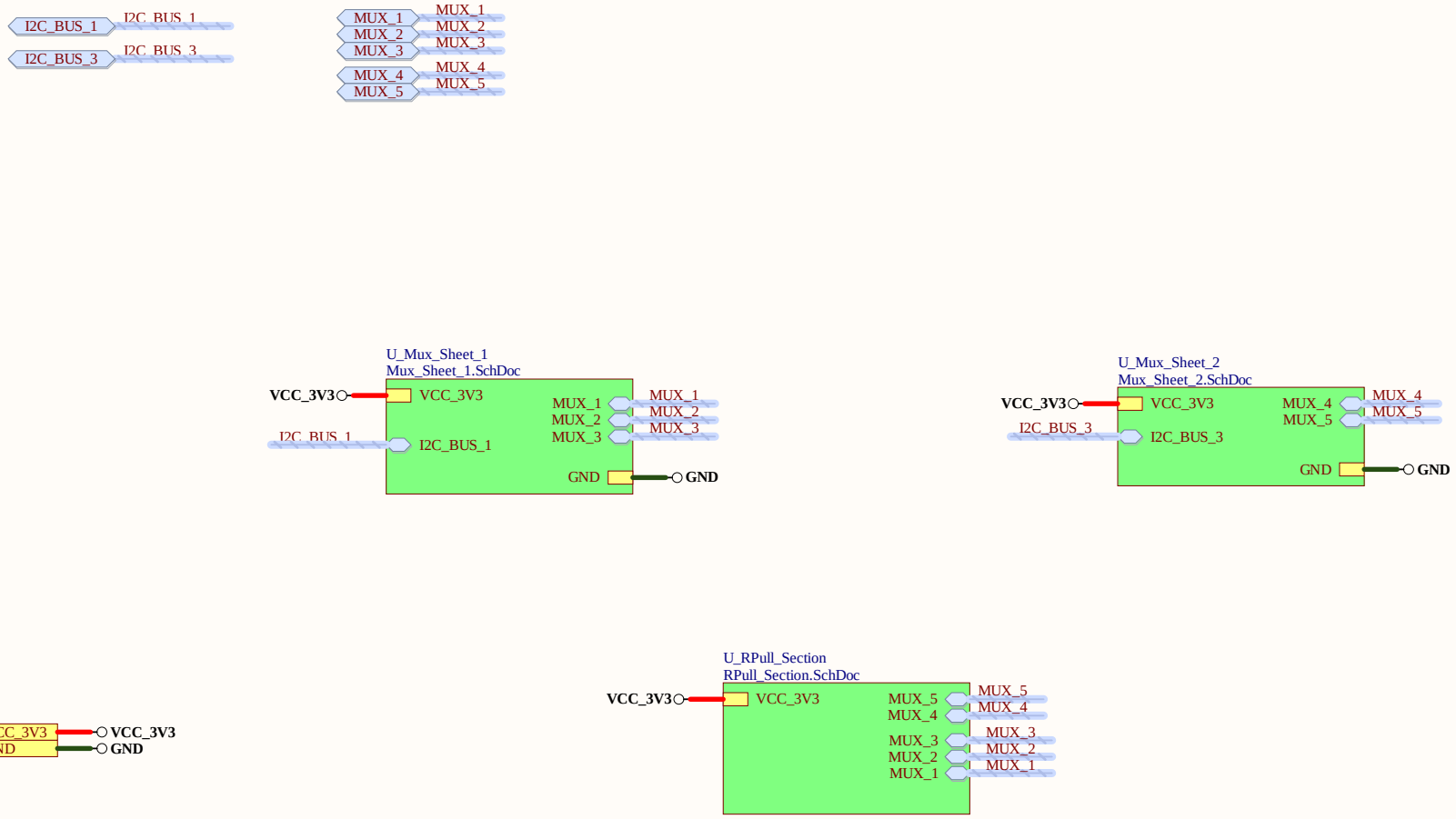

HSE & LSE CRYSTAL
XL_IN X1 32.768KHz XL_OUT
C16 10pF
C17 10pF
GND
GND
XH_IN X2 25MHz XH_OUT
C19 10pF
C20 10pF
GND
GND
GND

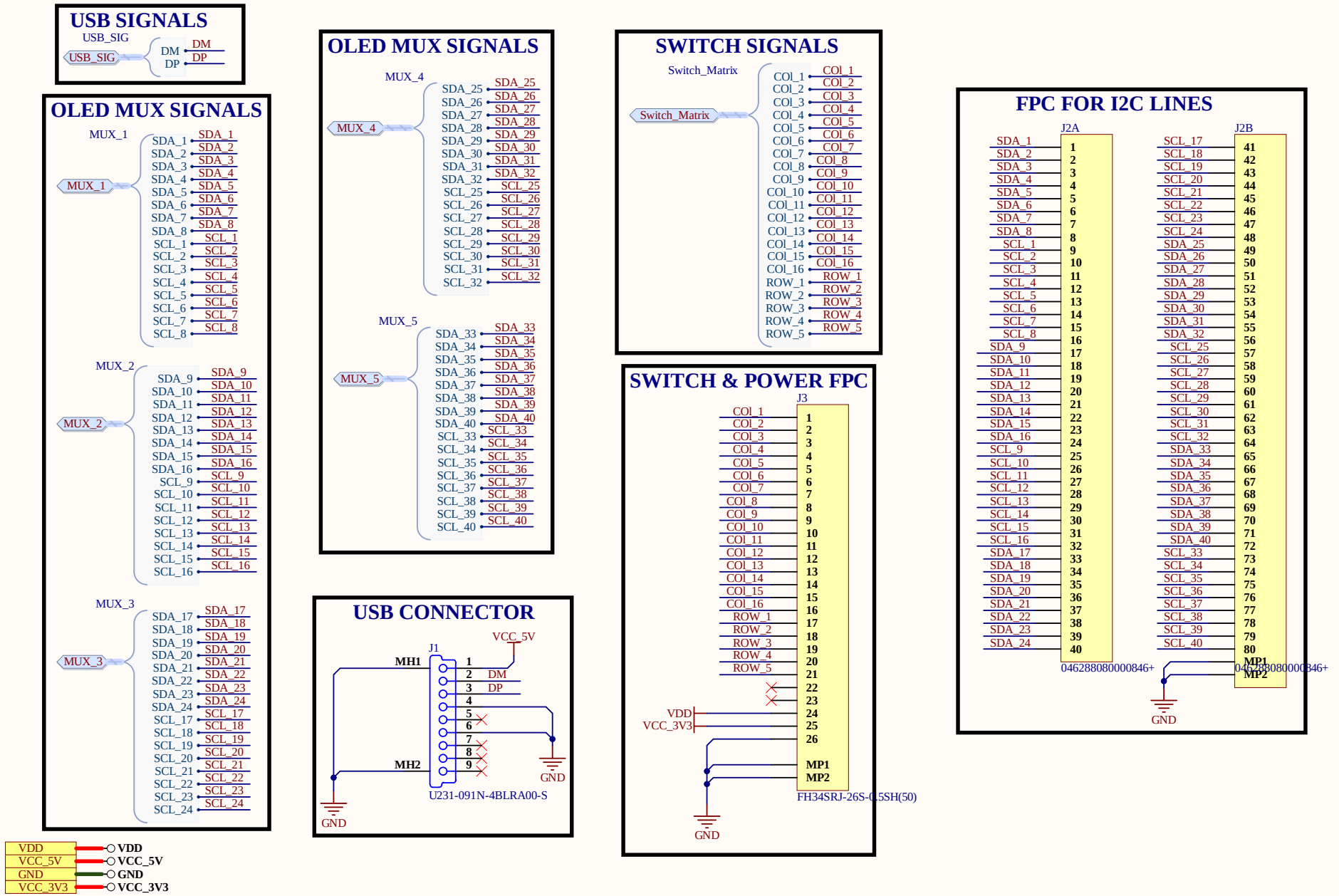

BOOT MODE SELECTOR
VCC_3V3
SW1
1
2 R2 10K BOOT0
3
SS-3235D-03-L2
GND
VCC_3V3
SW2
1
2 R3 10K BOOT1
3
SS-3235D-03-L2
GND

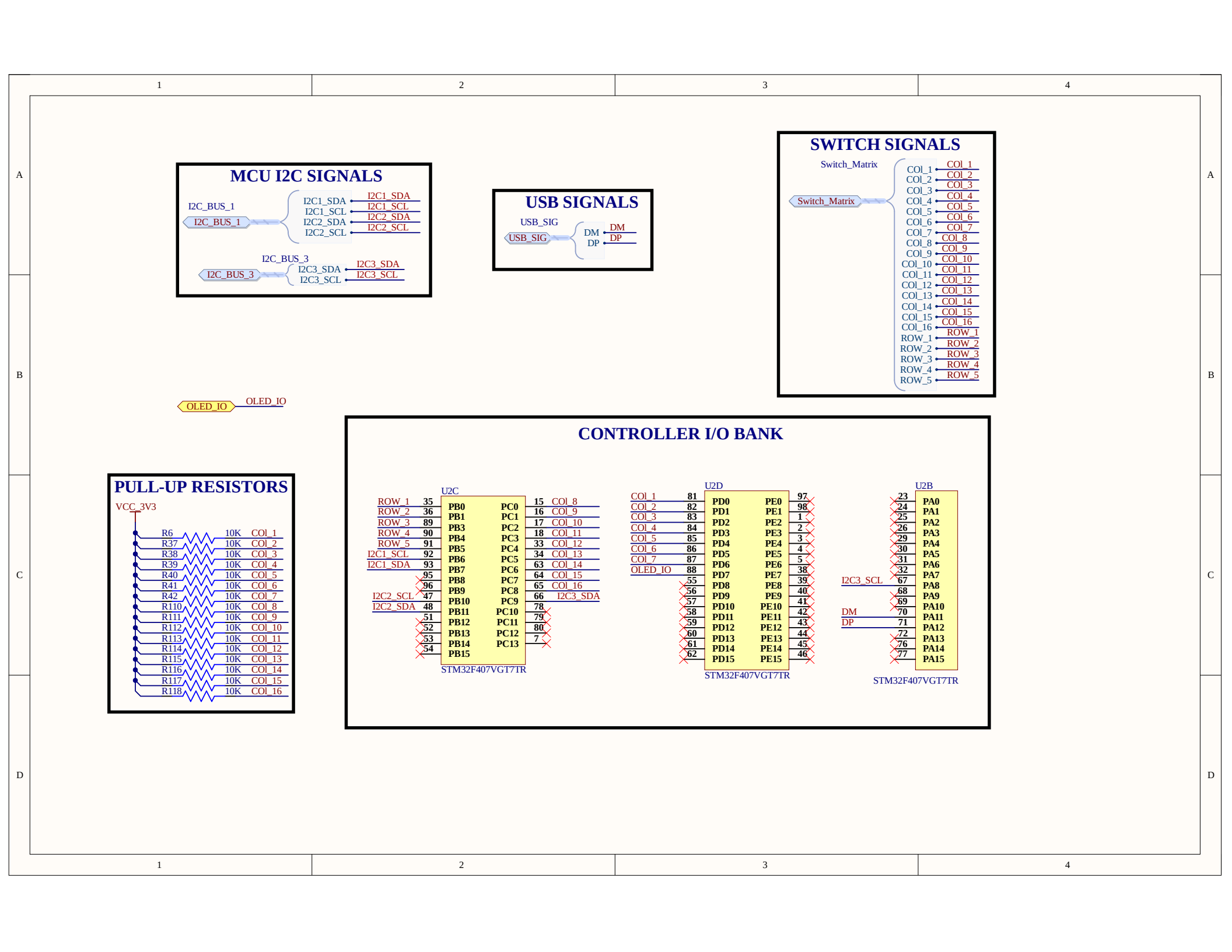

RESET SWITCH
NRST SW3
C18 0.1uF
GND

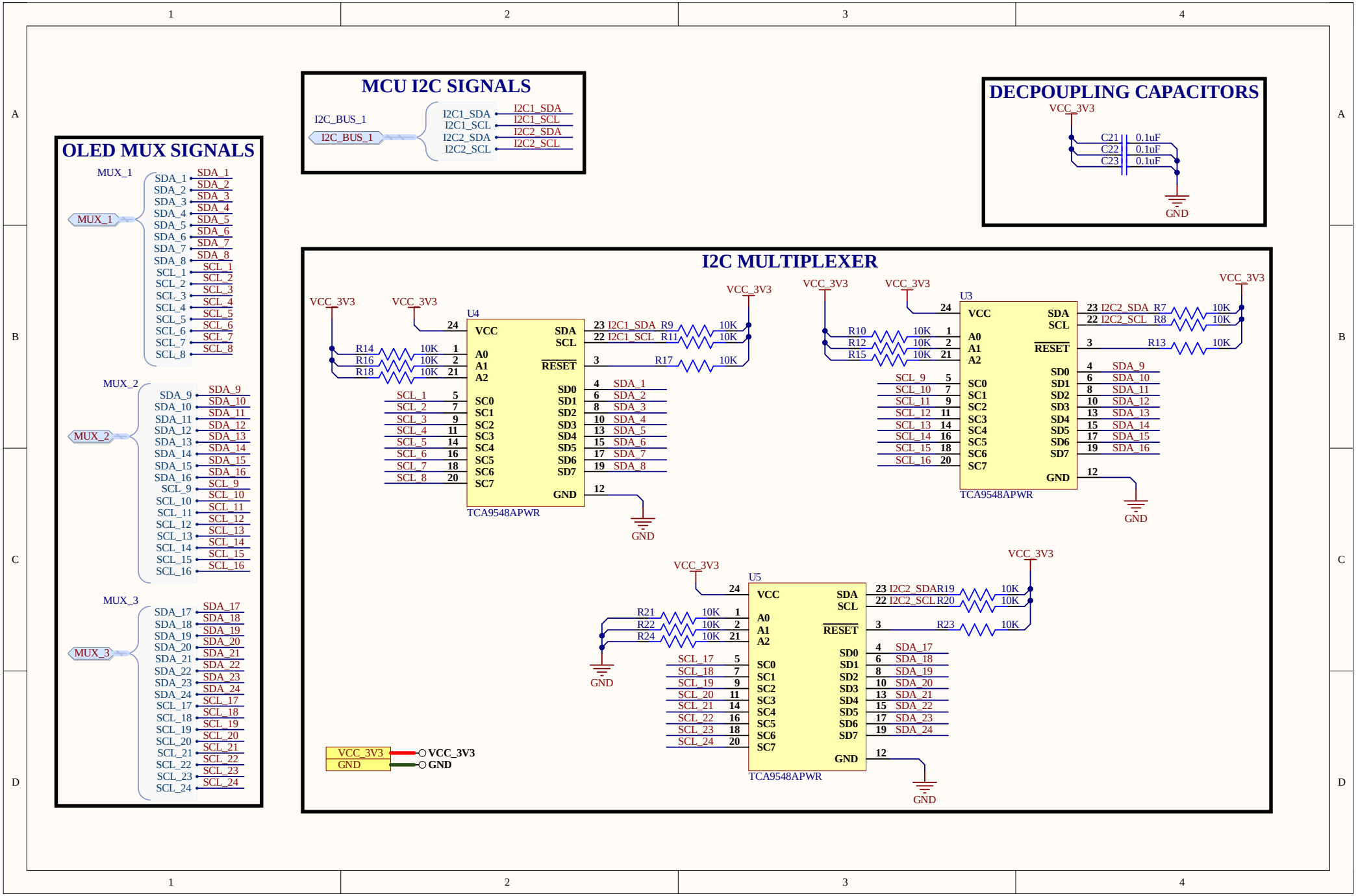

VCC_3V3
GND
VCC_3V3
GND

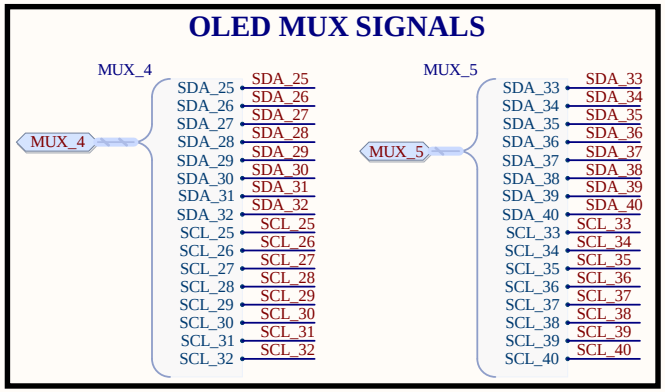

I2C_BUS_1
I2C_BUS_1
I2C_BUS_3
I2C_BUS_3
USB_SIG
USB_SIG
Switch_Matrix
Switch_Matrix
OLED_IO
OLED_IO
U_Controller_Section_1
Controller_Section_1.SchDoc
VCC_3V3
VCC_3V3
GND
GND
U_Controller_Section_2
Controller_Section_2.SchDoc
OLED_IO
OLED_IO
USB_SIG
USB_SIG
I2C_BUS_1
I2C_BUS_1
I2C_BUS_3
I2C_BUS_3
Switch_Matrix
Switch_Matrix
VCC_3V3
VCC_3V3
GND
GND

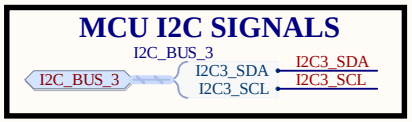

I2C_BUS_1
I2C_BUS_1
I2C_BUS_3
I2C_BUS_3
MUX_1
MUX_2
MUX_3
MUX_4
MUX_5
MUX_1
MUX_2
MUX_3
MUX_4
MUX_5
U_Mux_Sheet_1
Mux_Sheet_1.SchDoc
VCC_3V3
VCC_3V3
I2C_BUS_1
I2C_BUS_1
MUX_1
MUX_2
MUX_3
MUX_1
MUX_2
MUX_3
GND
GND
U_Mux_Sheet_2
Mux_Sheet_2.SchDoc
VCC_3V3
VCC_3V3
I2C_BUS_3
I2C_BUS_3
MUX_4
MUX_5
MUX_4
MUX_5
GND
GND
VCC_3V3
GND
VCC_3V3
GND
U_RPull_Section
RPull_Section.SchDoc
VCC_3V3
VCC_3V3
MUX_5
MUX_4
MUX_3
MUX_2
MUX_1
MUX_5
MUX_4
MUX_3
MUX_2
MUX_1

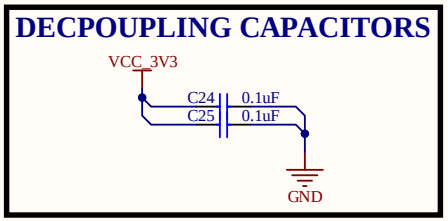

USB SIGNALS
USB_SIG
USB_SIG
DM
DP
DM
DP
OLED MUX SIGNALS
MUX_1
MUX_1
SDA_1 SDA_2 SDA_3 SDA_4 SDA_5 SDA_6 SDA_7 SDA_8
SCL_1 SCL_2 SCL_3 SCL_4 SCL_5 SCL_6 SCL_7 SCL_8
MUX_2
MUX_2
SDA_9 SDA_10 SDA_11 SDA_12 SDA_13 SDA_14 SDA_15 SDA_16
SCL_9 SCL_10 SCL_11 SCL_12 SCL_13 SCL_14 SCL_15 SCL_16
MUX_3
MUX_3
SDA_17 SDA_18 SDA_19 SDA_20 SDA_21 SDA_22 SDA_23 SDA_24
SCL_17 SCL_18 SCL_19 SCL_20 SCL_21 SCL_22 SCL_23 SCL_24
VDD
VCC_5V
GND
VCC_3V3
VDD
VCC_5V
GND
VCC_3V3
OLED MUX SIGNALS
MUX_4
MUX_4
SDA_25 SDA_26 SDA_27 SDA_28 SDA_29 SDA_30 SDA_31 SDA_32
SCL_25 SCL_26 SCL_27 SCL_28 SCL_29 SCL_30 SCL_31 SCL_32
MUX_5
MUX_5
SDA_33 SDA_34 SDA_35 SDA_36 SDA_37 SDA_38 SDA_39 SDA_40
SCL_33 SCL_34 SCL_35 SCL_36 SCL_37 SCL_38 SCL_39 SCL_40
USB CONNECTOR
VCC_5V
J1
MH1
MH2
1 2 3 4 5 6 7 8 9
DM
DP
GND
GND
U231-091N-4BLRA00-S
SWITCH SIGNALS
Switch_Matrix
Switch_Matrix
COl_1 COl_2 COl_3 COl_4 COl_5 COl_6 COl_7 COl_8 COl_9 COl_10 COl_11 COl_12 COl_13 COl_14 COl_15 COl_16
ROW_1 ROW_2 ROW_3 ROW_4 ROW_5
SWITCH & POWER FPC
J3
COl_1 COl_2 COl_3 COl_4 COl_5 COl_6 COl_7 COl_8 COl_9 COl_10 COl_11 COl_12 COl_13 COl_14 COl_15 COl_16
ROW_1 ROW_2 ROW_3 ROW_4 ROW_5
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26
MP1
MP2
VDD
VCC_3V3
GND
FH34SRJ-26S-0.5SH(50)
FPC FOR I2C LINES
J2A
J2B
SDA_1 SDA_2 SDA_3 SDA_4 SDA_5 SDA_6 SDA_7 SDA_8 SCL_1 SCL_2 SCL_3 SCL_4 SCL_5 SCL_6 SCL_7 SCL_8 SDA_9 SDA_10 SDA_11 SDA_12 SDA_13 SDA_14 SDA_15 SDA_16 SCL_9 SCL_10 SCL_11 SCL_12 SCL_13 SCL_14 SCL_15 SCL_16 SDA_17 SDA_18 SDA_19 SDA_20 SDA_21 SDA_22 SDA_23 SDA_24
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40
SCL_17 SCL_18 SCL_19 SCL_20 SCL_21 SCL_22 SCL_23 SCL_24 SDA_25 SDA_26 SDA_27 SDA_28 SDA_29 SDA_30 SDA_31 SDA_32 SCL_25 SCL_26 SCL_27 SCL_28 SCL_29 SCL_30 SCL_31 SCL_32 SDA_33 SDA_34 SDA_35 SDA_36 SDA_37 SDA_38 SDA_39 SDA_40 SCL_33 SCL_34 SCL_35 SCL_36 SCL_37 SCL_38 SCL_39 SCL_40
41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80
MP1
MP2
0462880800008 46+
0462880800008 46+
GND

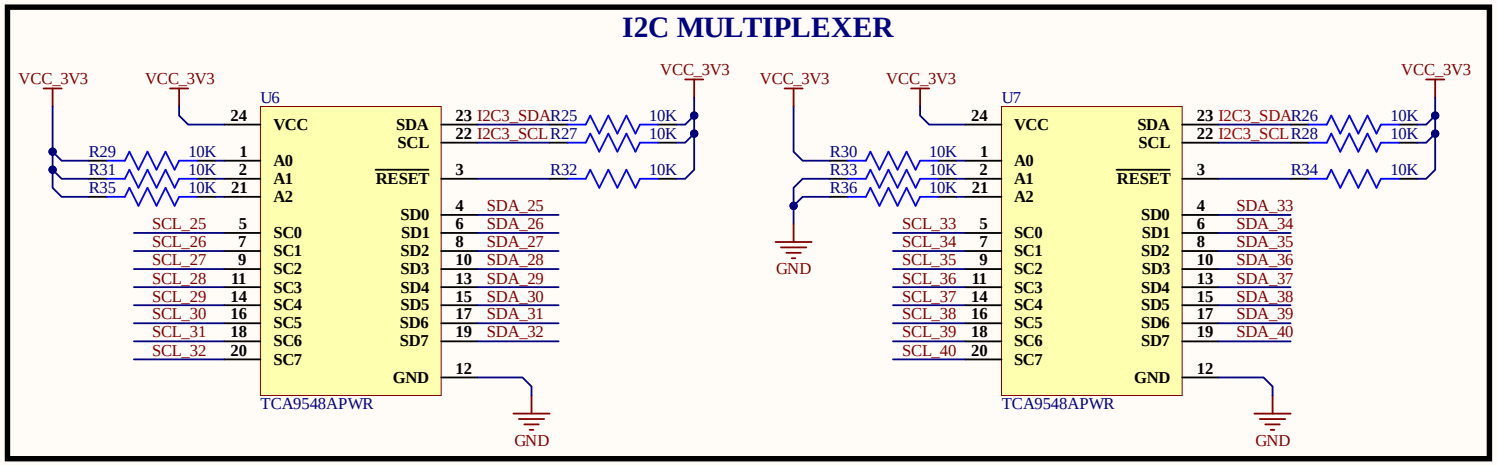

MCU I2C SIGNALS
I2C_BUS_1
I2C_BUS_1
I2C1_SDA
I2C1_SCL
I2C2_SDA
I2C2_SCL
I2C1_SDA
I2C1_SCL
I2C2_SDA
I2C2_SCL
I2C_BUS_3
I2C_BUS_3
I2C3_SDA
I2C3_SCL
I2C3_SDA
I2C3_SCL
USB SIGNALS
USB_SIG
USB_SIG
DM
DP
DM
DP
SWITCH SIGNALS
Switch_Matrix
Switch_Matrix
COl_1
COl_2
COl_3
COl_4
COl_5
COl_6
COl_7
COl_8
COl_9
COl_10
COl_11
COl_12
COl_13
COl_14
COl_15
COl_16
ROW_1
ROW_2
ROW_3
ROW_4
ROW_5
OLED_IO
OLED_IO
PULL-UP RESISTORS
VCC_3V3
R6 10K COl_1
R37 10K COl_2
R38 10K COl_3
R39 10K COl_4
R40 10K COl_5
R41 10K COl_6
R42 10K COl_7
R110 10K COl_8
R111 10K COl_9
R112 10K COl_10
R113 10K COl_11
R114 10K COl_12
R115 10K COl_13
R116 10K COl_14
R117 10K COl_15
R118 10K COl_16
CONTROLLER I/O BANK
U2C
ROW_1 35 PB0
ROW_2 36 PB1
ROW_3 89 PB3
ROW_4 90 PB4
ROW_5 91 PB5
I2C1_SCL 92 PB6
I2C1_SDA 93 PB7
95 PB8
96 PB9
I2C2_SCL 47 PB10
I2C2_SDA 48 PB11
51 PB12
52 PB13
53 PB14
54 PB15
PC0 15 COl_8
PC1 16 COl_9
PC2 17 COl_10
PC3 18 COl_11
PC4 33 COl_12
PC5 34 COl_13
PC6 63 COl_14
PC7 64 COl_15
PC8 65 COl_16
PC9 66 I2C3_SDA
PC10 78
PC11 79
PC12 80
PC13 7
STM32F407VGT7TR
U2D
COl_1 81 PD0
COl_2 82 PD1
COl_3 83 PD2
COl_4 84 PD3
COl_5 85 PD4
COl_6 86 PD5
COl_7 87 PD6
OLED_IO 88 PD7
55 PD8
56 PD9
57 PD10
58 PD11
59 PD12
60 PD13
61 PD14
62 PD15
PE0 97
PE1 98
PE2 1
PE3 2
PE4 3
PE5 4
PE6 5
PE7 38
PE8 39
PE9 40
PE10 41
PE11 42
PE12 43
PE13 44
PE14 45
PE15 46
STM32F407VGT7TR
U2B
23 PA0
24 PA1
25 PA2
26 PA3
29 PA4
30 PA5
31 PA6
32 PA7
I2C3_SCL 67 PA8
68 PA9
DM 69 PA10
DP 70 PA11
71 PA12
72 PA13
76 PA14
77 PA15
STM32F407VGT7TR


MCU I2C SIGNALS
I2C_BUS_1
I2C_BUS_1
I2C1_SDA
I2C1_SCL
I2C2_SDA
I2C2_SCL
I2C1_SDA
I2C1_SCL
I2C2_SDA
I2C2_SCL
DECPOUPLING CAPACITORS
VCC_3V3
C21 0.1uF
C22 0.1uF
C23 0.1uF
GND
OLED MUX SIGNALS
MUX_1
MUX_2
MUX_3
I2C MULTIPLEXER
U4
U3
U5
TCA9548APWR
VCC_3V3
GND
R14 10K
R16 10K
R18 10K
R17 10K
R10 10K
R12 10K
R15 10K
R21 10K
R22 10K
R24 10K
R9 10K
R11 10K
R8 10K
R7 10K
R13 10K
R19 10K
R20 10K
R23 10K


OLED MUX SIGNALS
MUX_4
MUX_4
SDA_25 SDA_25
SDA_26 SDA_26
SDA_27 SDA_27
SDA_28 SDA_28
SDA_29 SDA_29
SDA_30 SDA_30
SDA_31 SDA_31
SDA_32 SDA_32
SCL_25 SCL_25
SCL_26 SCL_26
SCL_27 SCL_27
SCL_28 SCL_28
SCL_29 SCL_29
SCL_30 SCL_30
SCL_31 SCL_31
SCL_32 SCL_32
MUX_5
MUX_5
SDA_33 SDA_33
SDA_34 SDA_34
SDA_35 SDA_35
SDA_36 SDA_36
SDA_37 SDA_37
SDA_38 SDA_38
SDA_39 SDA_39
SDA_40 SDA_40
SCL_33 SCL_33
SCL_34 SCL_34
SCL_35 SCL_35
SCL_36 SCL_36
SCL_37 SCL_37
SCL_38 SCL_38
SCL_39 SCL_39
SCL_40 SCL_40

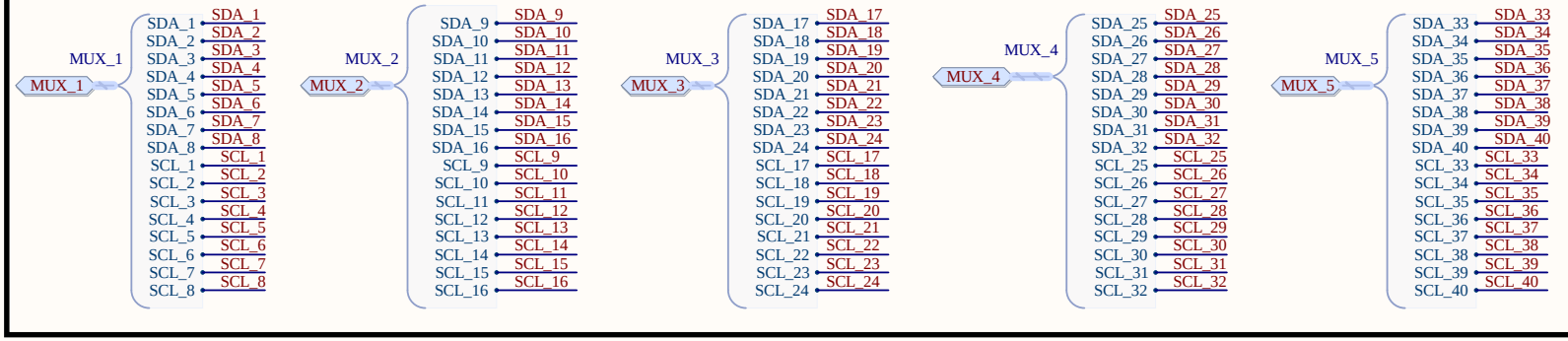

MCU I2C SIGNALS
I2C_BUS_3
I2C_BUS_3
I2C3_SDA
I2C3_SCL
I2C3_SDA
I2C3_SCL


DECPOUPLING CAPACITORS
VCC_3V3
C24 0.1uF
C25 0.1uF
GND

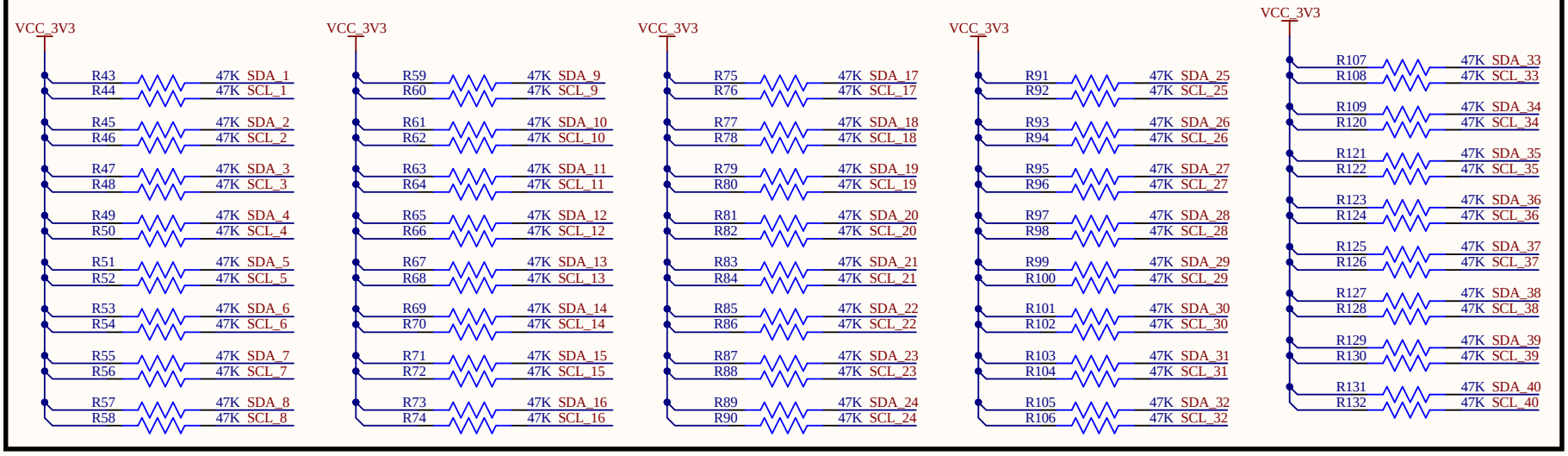

I2C MULTIPLEXER
VCC_3V3
VCC_3V3
U6
R29 10K
R31 10K
R35 10K
VCC
A0
A1
A2
SC0
SC1
SC2
SC3
SC4
SC5
SC6
SC7
SDA
SCL
RESET
SD0
SD1
SD2
SD3
SD4
SD5
SD6
SD7
GND
I2C3_SDA R25 10K
I2C3_SCL R27 10K
R32 10K
SCL_25 SDA_25
SCL_26 SDA_26
SCL_27 SDA_27
SCL_28 SDA_28
SCL_29 SDA_29
SCL_30 SDA_30
SCL_31 SDA_31
SCL_32 SDA_32
TCA9548APWR
GND
VCC_3V3
VCC_3V3
U7
R30 10K
R33 10K
R36 10K
I2C3_SDA R26 10K
I2C3_SCL R28 10K
R34 10K
SCL_33 SDA_33
SCL_34 SDA_34
SCL_35 SDA_35
SCL_36 SDA_36
SCL_37 SDA_37
SCL_38 SDA_38
SCL_39 SDA_39
SCL_40 SDA_40
TCA9548APWR
GND


VCC_3V3
GND
VCC_3V3
GND

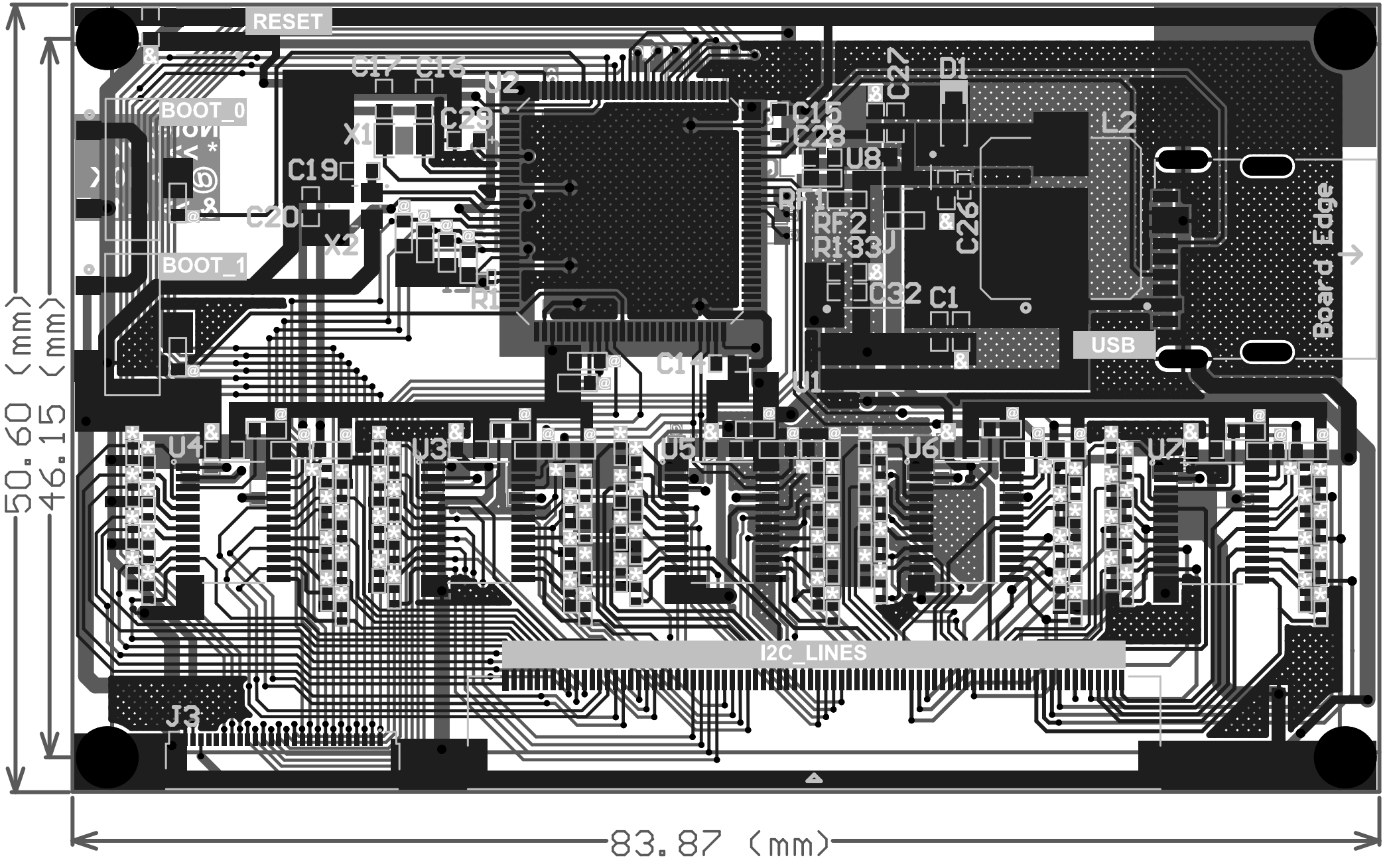
OLED MUX SIGNALS



PULL UP RESISTOR FOR I2C SIGNAL LINES






RESET
BOOT_0
BOOT_1
C17
C16
U2
X1
C29
C19
C20
X2
C15
C28
U8
C27
D1
L2
RF1
RF2
R133
C26
C32
C1
C11
USB
Board Edge
U1
U4
U3
U5
U6
U7
I2C_LINES
J3
50.60 (mm)
46.15 (mm)
83.87 (mm)